

THAI-HOC VU, Ph.D.

(name: THAI-HOC; surname: VU; born: 25/01/1995; nationality: VietNam)

Senior Postdoctoral Researcher | Wireless, AI & Quantum Communications

Faculty of Electrical Engineering and Computer Science
VSB–Technical University of Ostrava

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Research Profile

Postdoctoral researcher specializing in next-generation wireless and quantum communication systems, with expertise in performance analysis, mathematical optimization, and AI-assisted communication design. Research spans 6G wireless networks, reconfigurable intelligent surfaces, integrated sensing and communication, physical-layer security, quantum communications, and quantum key distribution. Author/co-author of 90+ publications with 2,200+ citations, including papers in leading IEEE Transactions and IEEE Communications Surveys & Tutorials. Current research focuses on bridging advanced wireless systems, AI, and quantum technologies for secure future communication networks.

Research at a Glance

90+	2,200+	25+	4	150+	3
Publications	Citations	h-index	Best Paper Awards	Verified Reviews	Grants

Research Expertise

Next-Generation Wireless Communication Systems	6G networks; RIS/STAR-RIS; RSMA/NOMA; short-packet communications; integrated sensing and communication; symbiotic communications.
Secure & Quantum Communications	Physical-layer security; covert communications; quantum key distribution; post-quantum security; quantum-assisted communications.
AI, Modeling & Optimization	Performance analysis; mathematical optimization; data-driven wireless design; generative AI; distributed learning.

Academic & Research Appointments

Postdoctoral Researcher (Full-Time)	Oct. 2025–Present
VSB–Technical University of Ostrava, Czech Republic	

- Conduct research on secure and intelligent next-generation wireless networks, including distributed reconfigurable intelligent systems and physical-layer security.
- Develop research directions integrating wireless and quantum technologies, including quantum key distribution, post-quantum security, quantum optimization, and AI-assisted quantum communications.
- Teach communication technologies and network modeling courses and supervise postgraduate research.

Lecturer (Remote)	Mar. 2025–Present
Institute of Digital Technology, Thu Dau Mot University, Vietnam	

- Contribute to postgraduate teaching, researcher training, and co-supervision of postgraduate researchers.

Postdoctoral Researcher	Apr. 2025–Sep. 2025
Kyonggi University, Republic of Korea	

- Developed new research directions and contributed to the planning of research projects.

Doctoral Researcher	Mar. 2020–Feb. 2025
University of Ulsan, Republic of Korea	

- Developed analytical and data-driven methodologies for performance analysis and optimization of next-generation wireless systems, including short-packet communications, NOMA/RSMA, RIS/STAR-RIS, energy harvesting, IoT, and physical-layer security.

- Published extensively in IEEE journals, developed international research collaborations, and co-supervised master's researchers.

Education

Ph.D. in Electrical, Electronic, and Information System Engineering (**Certified & Verified**) 2020–2025
University of Ulsan, Republic of Korea GPA: 4.06/4.5

Thesis: Improving Spectral and Energy Efficiency in Wireless Networks: From Analytical-Based Approaches to Deep Learning Designs ([Ulsan library](#))

Advisors: Prof. Sunghwan Kim and Assoc. Prof. Hee-Youl Kwak

B.Sc. in Electronics and Telecommunication Engineering (**Certified**) 2014–2018
Posts and Telecommunications Institute of Technology, Vietnam GPA: 2.94/4.0

Thesis: Study on Cooperative NOMA with Two-Stage Relay Selection

Advisors: Prof. Vo Nguyen Quoc Bao and Dr. Nguyen Duc Phuc

Research Funding & Projects

- **AI-Assisted Quantum Communications for Hybrid Wireless Networks**, IT4Innovations Open Access Grant Competition, OPEN-36-54, Role: Co-solver. [Project page](#) 2026-2027.
- **Networks and Communication Technologies for Smart Cities IX**, Ministry of Education, Youth and Sports, SP2026/004, Role: Co-solver. [Project page](#) 1/2026-12/2026.
- **Performance Enhancement of Integrated Sensing and Communication Networks Employing Federated Learning and Semantic Communications**, Vietnam National Foundation for Science and Technology Development (NAFOSTED), 102.02-2025.92, Role: Member. [Funding decision](#) 2026-2028.
- **Improving Private Communications for Next-Gen by Exploiting Advanced Multiple Access and Reconfigurable Surface**, NAFOSTED, 102.02-2025.97, Role: Member. [Funding decision](#) 2026-2028.

Selected Awards & Honours

- **Vietnam Golden Globe Award in Science and Technology for Young Researchers**, 2025 (**Certificate**).
- **World Top 2% Scientists**, Stanford/Elsevier Single-Year Impact, 2025 (**Certificate**).
- **Best Paper Awards:** Radioelektronika 2026 (**Certificate**); IEEE ATC 2024 (**Certificate**); IEEE ICCE 2024 and 2022 (**Certificate**).
- **IEEE Communications Letters Exemplary Reviewer**, 2023 (**Certificate**) and 2024 (**Webpage Recognition**).
- **Excellent SCI(E) Paper, BK21 Program**, five awards, 2021–2025.
- **Vietnamese Young Scientists in Korea Award**, 2022 (**Certificate**).
- **Student research excellence**, PTIT, Vietnam, January 2018 (**Certificate**).

Research Supervision & Mentoring

Main Supervisor | Master's Research, VSB–Technical University of Ostrava

Thesis: *Design and Development of Quantum Key Distribution over Satellite Networks*

- Guide research-problem formulation, methodological design, system modeling, simulation, and performance evaluation.
- Mentor the student in literature analysis, scientific reasoning, technical writing, and presentation of research findings.
- Monitor research milestones and provide regular scientific and technical feedback toward independent research.

Co-Supervisor | Ph.D. & Master's Research, Vietnam

Currently co-supervising **01** Ph.D. researcher at Posts and Telecommunications Institute of Technology and **01** master's researcher at Thu Dau Mot University, with responsibilities spanning research scope, methodology, technical analysis, progress monitoring, and publication development.

Research Mentor / Co-Supervisor | Master's Research, University of Ulsan

Co-supervised two master's researchers in RIS/STAR-RIS, short-packet communications, RSMA, deep learning, and covert communications. Their research contributed to publications in *IEEE Transactions on Vehicular Technology* (**Records:** [1], [2]), *IEEE Internet of Things Journal* (**Records:** [3]), *IEEE Access* (**Records:** [4]), and awards-winning IEEE ICCE conference paper (**Certificate**) and IEEE ATC 2024 conference paper (**Certificate**).

Teaching Experience

Lecturer, 440-4107/01 Diploma Project I 2026

VSB–Technical University of Ostrava

Understand given the task in the field of communication and information technology and solutions to manage team.

Lecturer, 440-4218/02 Modeling and Dimensioning of Networks 2026

VSB–Technical University of Ostrava

Network modeling and dimensioning; queueing systems; GNS3, OMNeT++, ns-3; LPWAN modeling and simulation.

Co-Lecturer, 440-2103/04 Introduction to Communication Technologies 2025

VSB–Technical University of Ostrava

Communication networks; optical and wireless systems; mobile technologies; IoT; communication security.

Co-Lecturer, 8HTTT006 Advanced Data Mining 2025

Thu Dau Mot University

Data preprocessing; data-mining algorithms; computational methods and data-driven analysis.

Selected Publications

Representative selection. For the complete publication record, see [Google Scholar](#).

Survey/Review articles

1. **T.-H. Vu**, N. H. Tu, et al., “*Integration of TinyML and LargeML: A survey of 6G and beyond*,” *IEEE Internet of Things Journal*, 2026. (15+ citations)
2. N. C. Luong, T. Huynh-The, **T.-H. Vu**, et al., “*Advanced Learning Algorithms for Integrated Sensing and Communication (ISAC) Systems in 6G and Beyond: A Comprehensive Survey*,” *IEEE Communications Surveys and Tutorials*, 2025. (70+ citations)
3. O. Aouedi, **T.-H. Vu**, et al., “*A Survey on Intelligent Internet of Things: Applications, Security, Privacy, and Future Directions*,” *IEEE Communications Surveys and Tutorials*, 2025. (450+ citations)
4. M. Le, H.-T. Thien, D.-D. Tan, **T.-H. Vu**, et al., “*Applications of Distributed Machine Learning for the Internet-of-Things: A Comprehensive Survey*,” *IEEE Communications Surveys and Tutorials*, 2025. (110+ citations)
5. **T.-H. Vu**, SK Jagatheesaperumal, MD Nguyen, et al., “*Applications of Generative AI (GAI) for Mobile and Wireless Networking: A Survey*,” *IEEE Internet of Things Journal*, 2025. (110+ citations)

Magazine articles

1. **T.-H. Vu**, M. Zeng, S. Kim, H. V. Poor, and Q.-V. Pham, “*Short-Packet Communications: Recent Advances and Research Challenges*,” *IEEE Network*, 2025.

Technical articles

Top Journal

1. **T.-H. Vu**, D. B. da Costa, S. Kim, and Q.-V. Pham, “*Outage, Capacity, and Error Performance of Downlink RSMA-Based Systems: Analysis and Resource Optimization*,” *IEEE Transactions on Communications*, 2025. (10+ citations)
2. **T.-H. Vu**, T.-V. Nguyen, D. B. da Costa, and S. Kim, “*Reconfigurable Intelligent Surface-Aided Cognitive NOMA Networks: Performance Analysis and Deep Learning Evaluation*,” *IEEE Transactions on Wireless Communications*, 2022. (40+ citations)
3. **T.-H. Vu**, T.-V. Nguyen, T.-T. Nguyen, and S. Kim, “*Performance Analysis and Deep Learning Design of Wireless Powered Cognitive NOMA IoT Short-Packet Communications With Imperfect CSI and SIC*,” *IEEE Internet of Things Journal*, 2022. (80+ citations)

4. **T.-H. Vu**, T.-V. Nguyen, D. B. da Costa, and S. Kim, “*Intelligent reflecting surface-aided short-packet non-orthogonal multiple access systems*,” *IEEE Transactions on Vehicular Technology*, 2022. (99+ citations)
5. **T.-H. Vu**, T.-V. Nguyen, Q.-V. Pham, D. B. da Costa, and S. Kim, “*Short-packet communications for UAV-based NOMA systems under imperfect CSI and SIC*,” *IEEE Transactions on Cognitive Communications and Networking*, 2022. (50+ citations)

Academic Leadership & Professional Service

- **Special Session Chair**, International Conference On Computational Intelligence In Engineering Science (ICCIES) 2027 ([Webpage](#))
- **Technical Program Committee Member**, RIVF - 20th International Conference on Computing and Communication Technologies, 2026 ([Webpage](#)).
- **Technical Program Committee Member & Session Chair**, 36th International Conference Radioelektronika, Ostrava, Czech Republic, 2026 ([Webpage](#)).
- **Special Session Chair**, International Conference on Advanced Technologies for Communications (ATC), 2025 ([Webpage](#)).
- **Technical Program Committee Member**, International Conference on Research in Intelligent Computing in Engineering (RICE), 2022.
- **Peer Review**: 150+ manuscript reviews, verified by Web of Science ([Webpage](#)); IEEE Communications Letters Exemplary Reviewer in 2023 and 2024 ([Webpage Recognition](#)).

Invited Talks & Outreach

- Phenikaa University: research orientation, effective learning, and research-career experience sharing ([Video Record](#)).
- Vietnamese Students Association: student scientific conference on 5G/6G mobile network technology, 17 Nov. 2025.

Earlier Industry Experience

R&D Engineer Jan. 2019–Jan. 2020

Saigontourist Cable Television, Vietnam

Worked on HFC/GPON network design, **operational systems, infrastructure troubleshooting, and technical appraisal**.

Embedded IoT Engineer Intern Jul. 2018–Nov. 2019

TMA Solutions, Vietnam

Developed IoT connectivity and AWS LoRa gateway solutions using embedded devices, APIs, databases, and web interfaces (**Full stack developments**).

Academic References

1. **Prof. Miroslav Voznak** (Head of Research Group), Faculty of Electrical Engineering and Computer Science, VSB-Technical University of Ostrava, Ostrava, Czechia - E-mail: miroslav.voznak@vsb.cz. [Google Scholar](#).
2. **Prof. SungHwan Kim** (Doctoral Advisor), Full-Professor - School of Electronic Engineering, Kyonggi University, Suwon, South Korea - E-mail: skim@kyonggi.ac.kr. [Google Scholar](#).
3. **Prof. Daniel Benevides da Costa** (Collaborator), Distinguished University Professor - Department of Electrical Engineering, King Fahd University of Petroleum & Minerals (KFUPM), Dhahran 31261, Saudi Arabia; EiC of IEEE COMML - Email: danielbcosta@ieee.org. [Google Scholar](#).
4. **Prof. Viet Quoc Pham** (Collaborator), Assistant Professor - School of Computer Science and Statistics, Trinity College Dublin, Dublin 2, D02 PN40, Ireland - E-mail: viet.pham@tcd.ie. [Google Scholar](#).
5. **Prof. Vo Nguyen Quoc Bao** (Bachelor Advisor), Full Professor, Vice Rector - Van Lang School of Technology, Van Lang University, Vietnam - Email: bao.vnq@vlu.edu.vn. [Google Scholar](#).